Thermodynamic description of the Cu–Ni system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cu–Ni binary system as calculated with pycalphad from the SGTE dataset CuNi-87Jan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key.

1. Introduction

The Cu–Ni system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as isomorphous, solid miscibility gap. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of A. Jansson, TRITA-MAC 340, KTH, Stockholm 1987. [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Cu, the liquidus meets the axis at 1357.8 K. For Ni, the liquidus meets the axis at 1728.2 K.

2. Phase diagram and invariant reactions

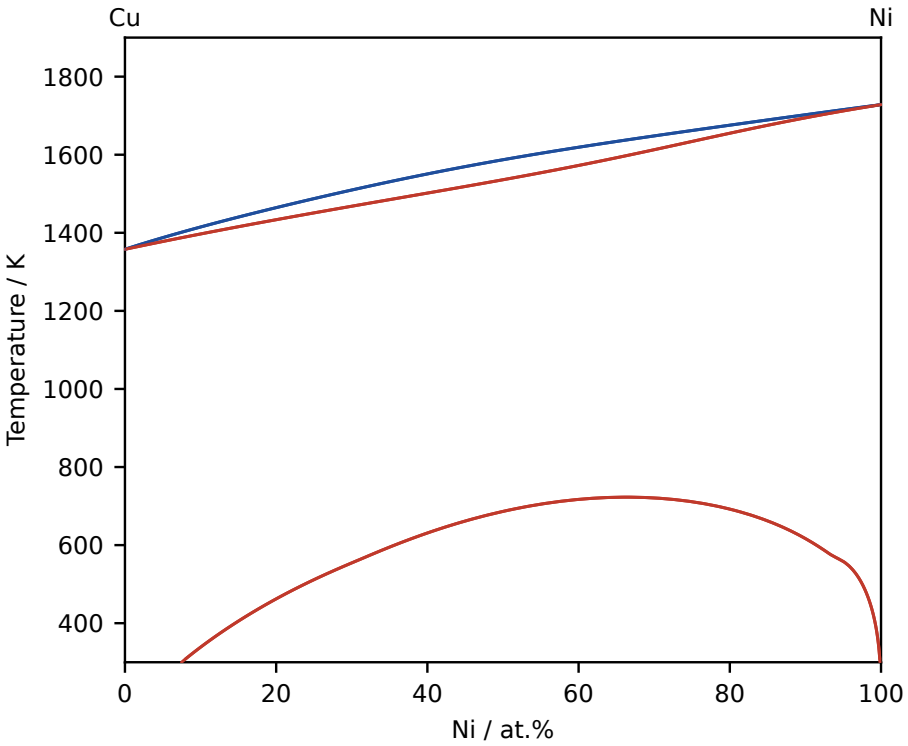


Fig. 1. Calculated Cu–Ni phase diagram.

No invariant reaction occurs between 300 K and the melting points; Cu and Ni form a continuous solid solution at high temperature.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Cu–Ni).

Reaction	Type	T / K		Compositions (at.% Ni)		
		this work	exp.[b]	phase 1	phase 2	phase 3
L → (Cu,Ni)	melting (pure component)[a]	1728.2	1730.2 [4]	100.0	100.0	–
L → (Cu,Ni)	melting (pure component)[a]	1357.8	1359.8 [5]	0.0	0.0	–

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

[a] Two-phase equilibrium at a temperature extremum; both phases have the same composition.

[b] Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

Table 2. Liquidus and solidus temperatures (calculated) (Cu–Ni).

at.% Ni	T liquidus / K	T solidus / K
20	1464.4	1433.5
40	1550.7	1501.7
60	1619.1	1572.5
80	1675.6	1655.0

3. Phase stability

Table 3. Solid phases: structure, stability range and homogeneity range (Cu–Ni).

	Structure			Stability range / K		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Ni)	at T / K
(Cu,Ni)	cF4	A1	Cu	< 300	1728.2	0.000–1.000	430

The upper stability limit of (Cu,Ni) is 1728.2 K.

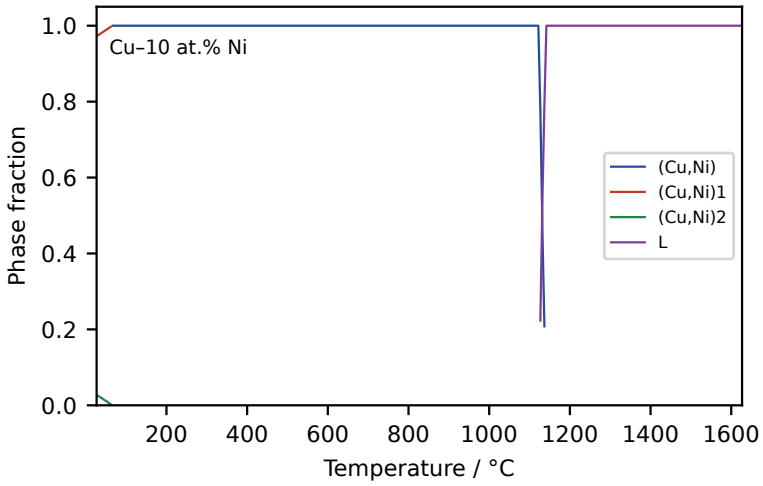


Fig. 2. Calculated equilibrium phase fractions versus temperature for Cu–10 at.% Ni.

4. Thermodynamic properties

At 1800 K the integral mixing enthalpy of the liquid has its maximum, 2954 J/mol, at 54.0 at.% Ni. For the equiatomic liquid at 1800 K, the Gibbs energy of mixing is −6.95 kJ/mol-atom.

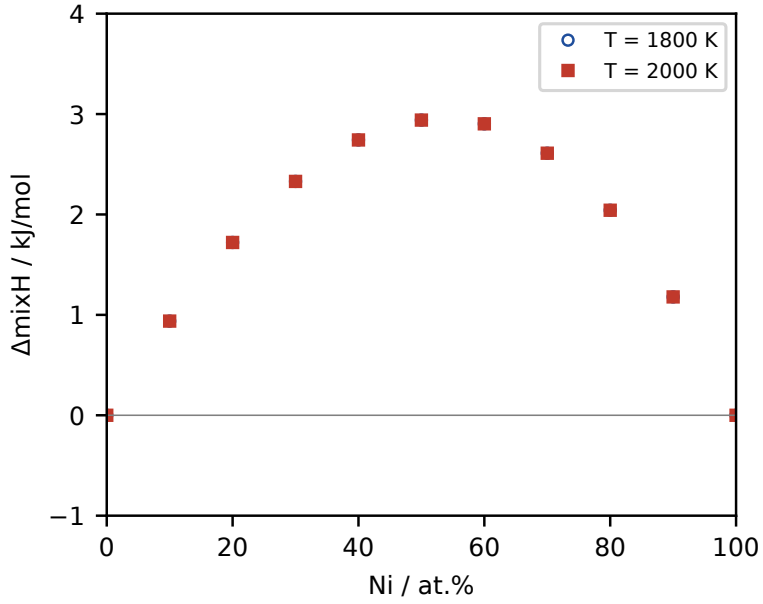


Fig. 3. Calculated mixing enthalpy of the liquid at 1800 K and 2000 K.

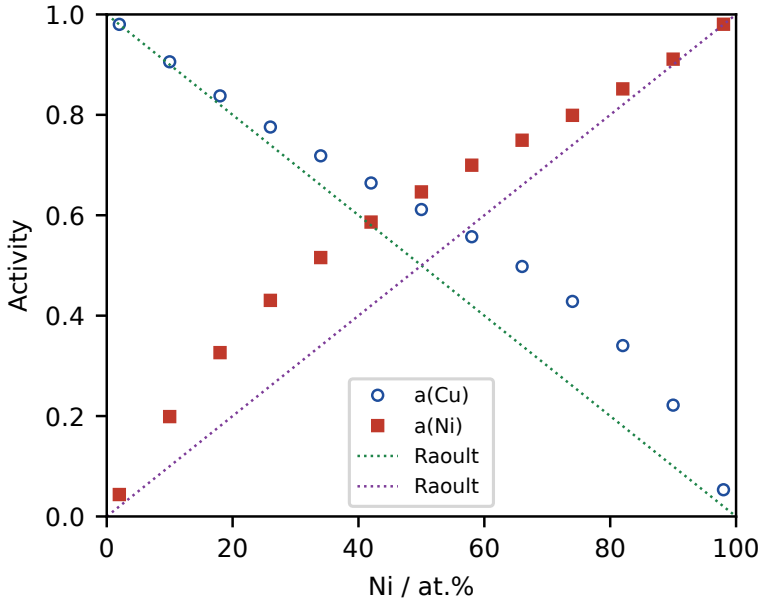


Fig. 4. Calculated activities in the liquid at 1800 K (reference: pure liquids).

Table 4. Redlich–Kister parameters of the liquid (Cu–Ni).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	11760.0	1.084	0.0
L1	-1671.8	0.0	0.0

Table 5. Integral and partial mixing properties of the liquid at 1800 K (Cu–Ni).

at.% Ni	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Cu)	a(Ni)
10	0.94	-3.75	0.9057	0.1989
30	2.33	-6.40	0.7465	0.4752
50	2.94	-6.95	0.6114	0.6465
70	2.61	-6.12	0.4649	0.7740
90	1.18	-3.51	0.2217	0.9109

Reference states: pure liquid Cu and Ni at 1800 K; per mole of atoms.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] A. Jansson, TRITA-MAC 340, KTH, Stockholm 1987.

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.